Bahdleen VPN

OpenVPN + WireGuard Installer & Manager

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This document showcases the installation and usage flow of **Bahdleen VPN**, a personal portfolio project focused on delivering a reliable, user-friendly OpenVPN and WireGuard deployment experience across major Linux distributions.

Contents

1 What This Project Covers

Bahdleen VPN focuses on:

- Cross-distro installation logic for OpenVPN and WireGuard.
- Automated OpenVPN PKI setup (Easy-RSA) and TLS material generation.
- Guided public IP detection with manual fallback.
- DNS selection to push to clients.
- NAT interface detection and persistence.
- Client and peer lifecycle management.
- Export of configs to user-owned directories for clarity and portability.

2 Current Boundaries

This version does not aim to be an enterprise HA platform. It prioritizes a reliable single-server workflow suitable for: portfolio demonstration, lab environments, and production-style proof-of-concepts.

In some environments:

- Public IP lookup can be blocked by provider security.
- Custom firewall baselines may require additional manual rules.
- HA clustering and multi-region failover remain future enhancements.

3 Installation & Usage Walkthrough

This section documents the real installation and setup flow of **Bahdleen VPN** as executed in a live environment. The screenshots are presented in sequence to demonstrate usability, automation logic, and the full deployment lifecycle from first run to client export.

3.1 Getting the Script Ready

The setup begins by confirming the script is present in the working directory, making it executable, and running it with elevated privileges. This ensures the installer can manage packages, services, firewall rules, and VPN interfaces without requiring repeated manual intervention.

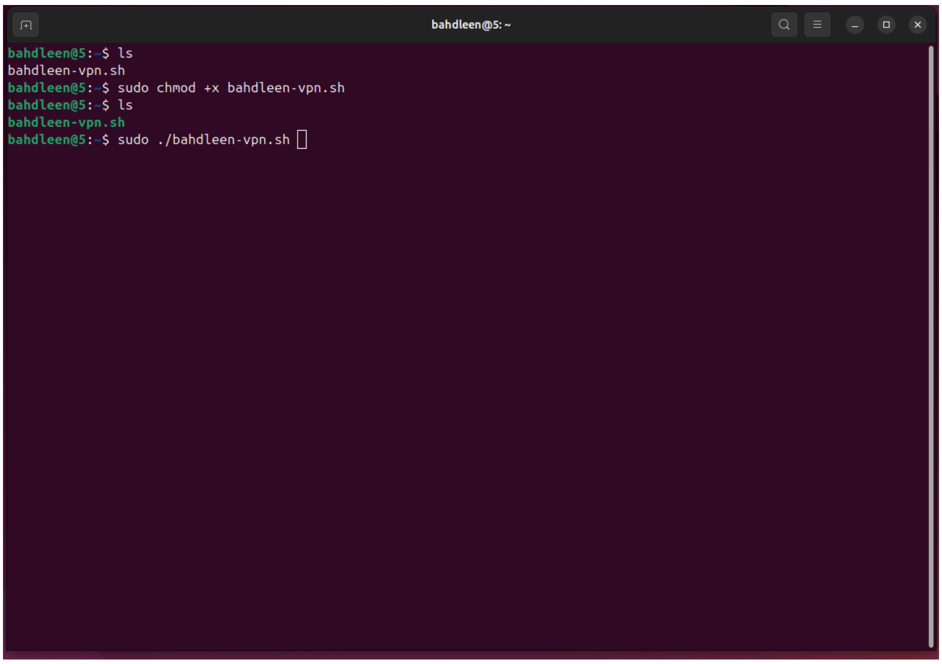

Figure 1: Listing the project directory to confirm the presence of `bahdleen-vpn.sh`.

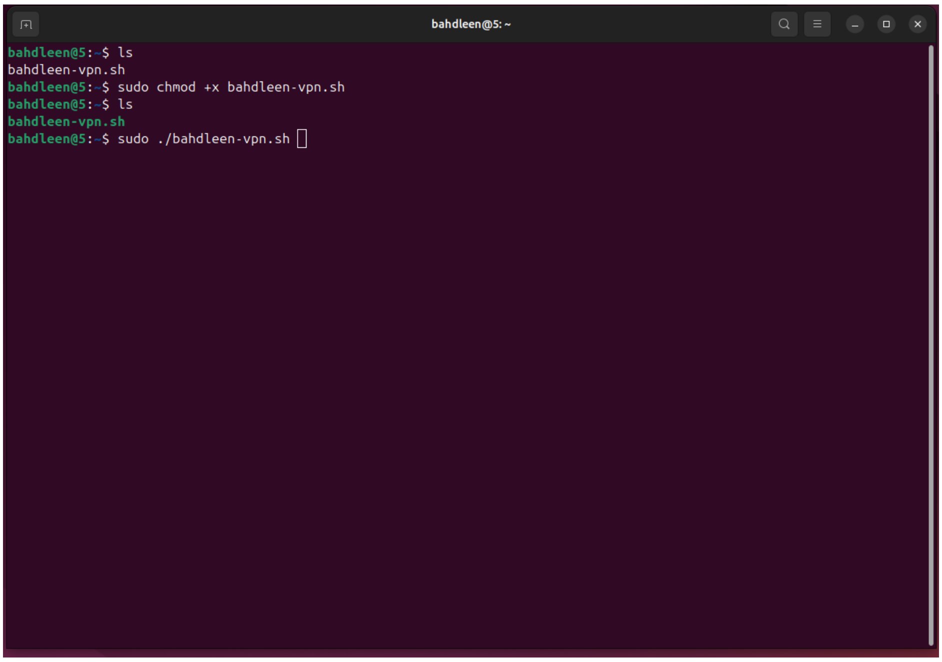

Figure 2: Granting execute permission using `chmod +x`.

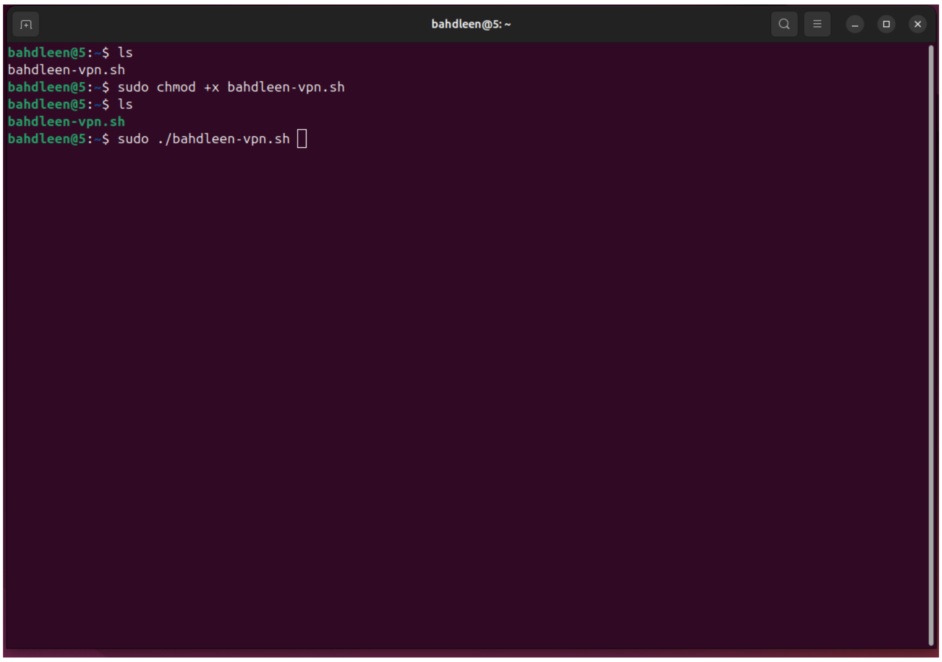

Figure 3: Launching the script with `sudo ./bahdleen-vpn.sh`.

3.2 Menu Interface and Export Paths

On launch, Bahdleen VPN displays a branded ASCII header alongside system context, including detected OS, active user, and the export destinations for client files. A key design choice is keeping client profiles in user-owned directories rather than hidden system paths.

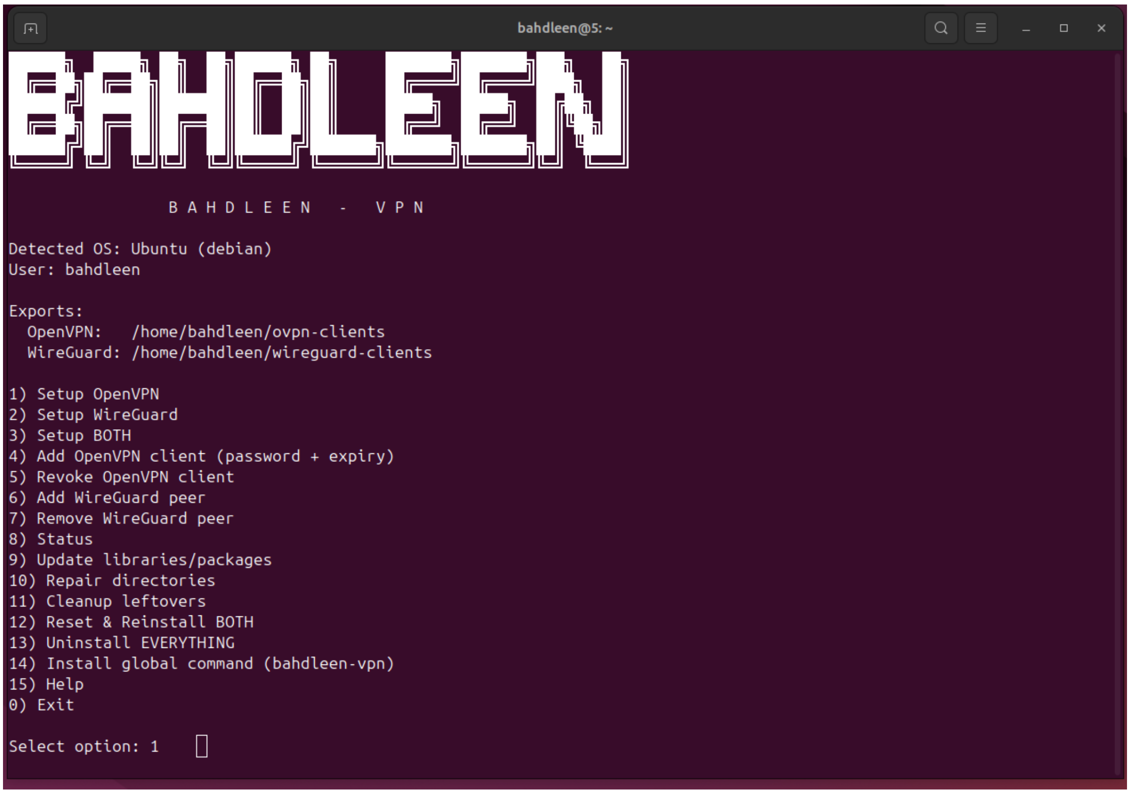

Figure 4: The Bahdleen VPN main menu showing detected OS, user, and export locations.

The home-based export model improves portability and reduces confusion during testing, demos, and documentation: `~/ovpn-clients` for OpenVPN and `~/wireguard-clients` for WireGuard.

3.3 Installing the Global Command

A major usability feature is the ability to install a global launcher so the tool can be reopened later using a single command: `bahdleen-vpn`. This also includes a carefully scoped passwordless sudo rule restricted only to this tool path, which prevents broad escalation while improving user experience.

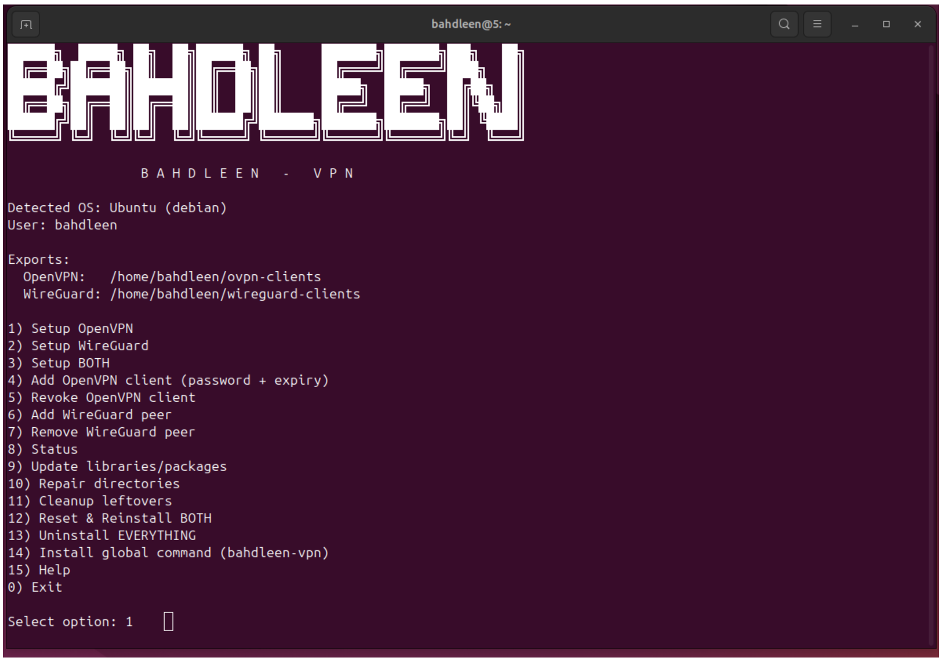

Figure 5: Global command install prompt showing the script path, launcher path, and restricted sudo policy.

3.4 Running a Combined Deployment (OpenVPN + WireGuard)

The tool supports independent or combined installation. In this walkthrough, the combined option is used to validate the full automation workflow and NAT consistency across both VPN services.

Figure 6: Selecting the combined setup option for OpenVPN + WireGuard.

3.5 Static Private IP Advisory

Before deployment proceeds, the script includes a short private/LAN IP advisory. This does not force configuration changes but promotes best practice by warning that VPN stability is improved when the server's private IP is static.

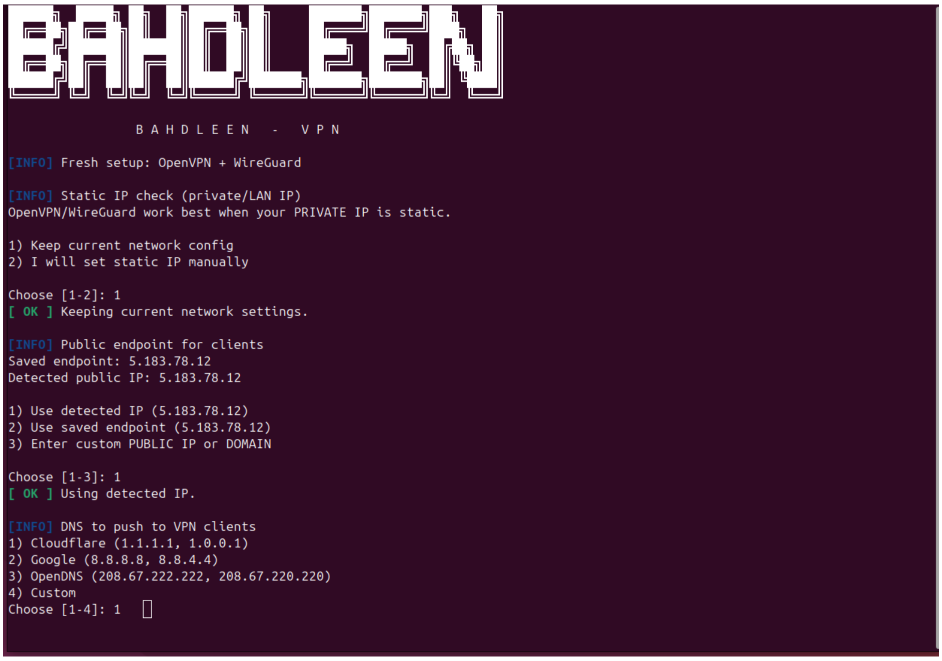

Figure 7: Private/LAN IP advisory with options to keep current settings or set static manually.

3.6 Public Endpoint Detection with Safe Fallback

A common issue in many VPN installers is unreliable or blocked public IP detection. Bahdleen VPN implements a resilient approach: it detects the public IP when possible, shows it to the user, and also offers a saved endpoint or manual input option.

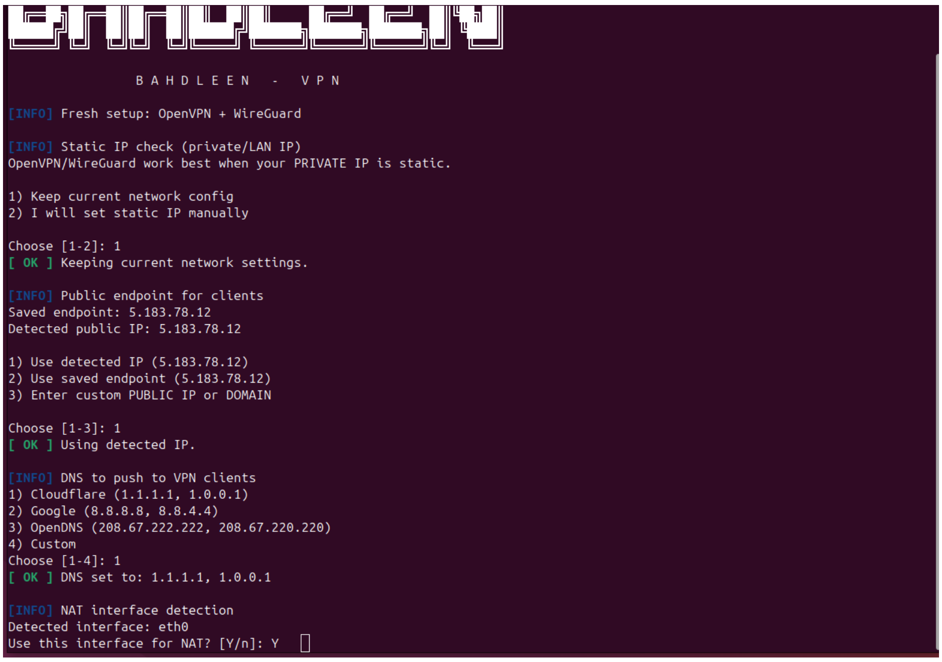

Figure 8: Public IP detection flow showing saved endpoint and detected endpoint options.

3.7 DNS Selection for Clients

To keep the deployment suitable for real usage, the script prompts for DNS preferences that will be pushed to VPN clients. This provides a clear balance between convenience and control.

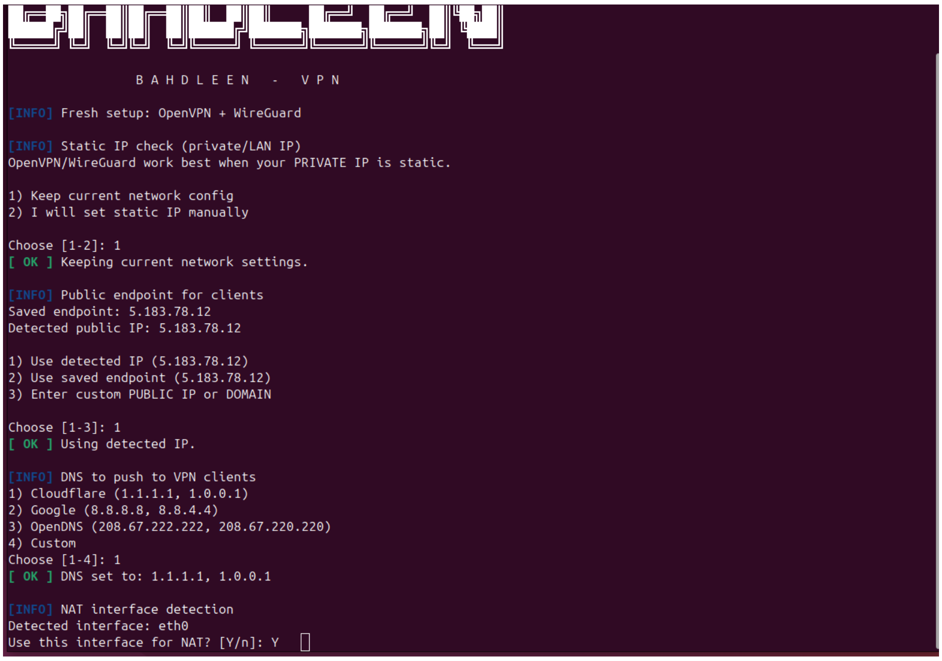

Figure 9: DNS selection menu for client-side configuration.

3.8 NAT Interface Detection

Because NAT is required for common remote-access VPN scenarios, the tool detects the primary egress interface and asks for confirmation before applying persistent rules.

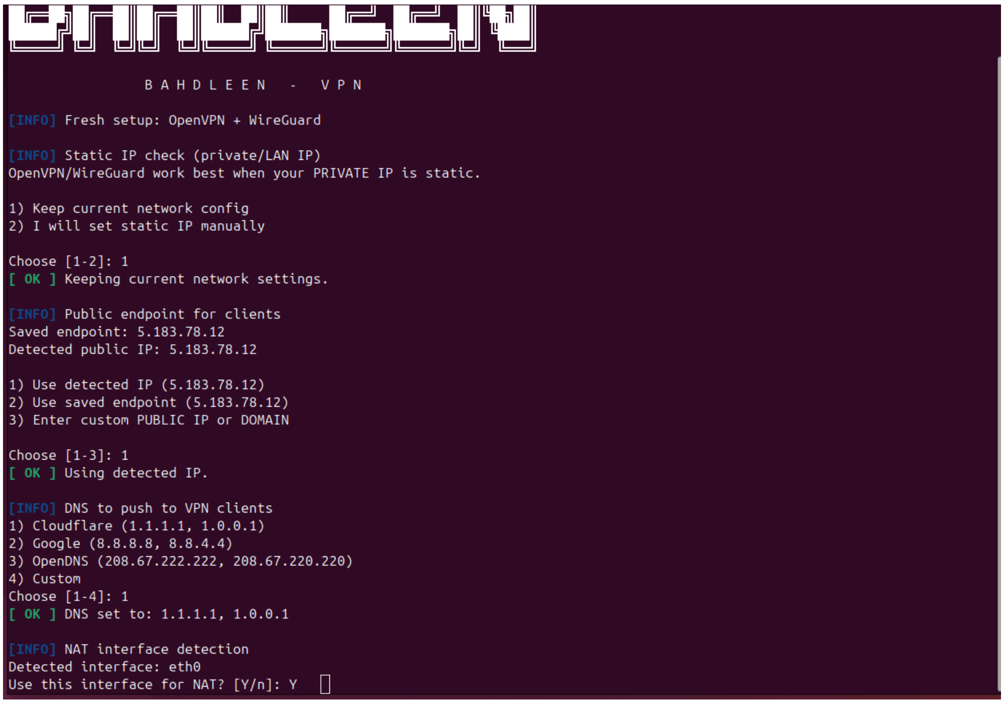

Figure 10: NAT interface detection with confirmation prompt.

3.9 OpenVPN PKI and Cryptographic Setup

The OpenVPN setup process includes fully automated PKI initialization using Easy-RSA. This covers server certificates, TLS material, and Diffie-Hellman parameters.

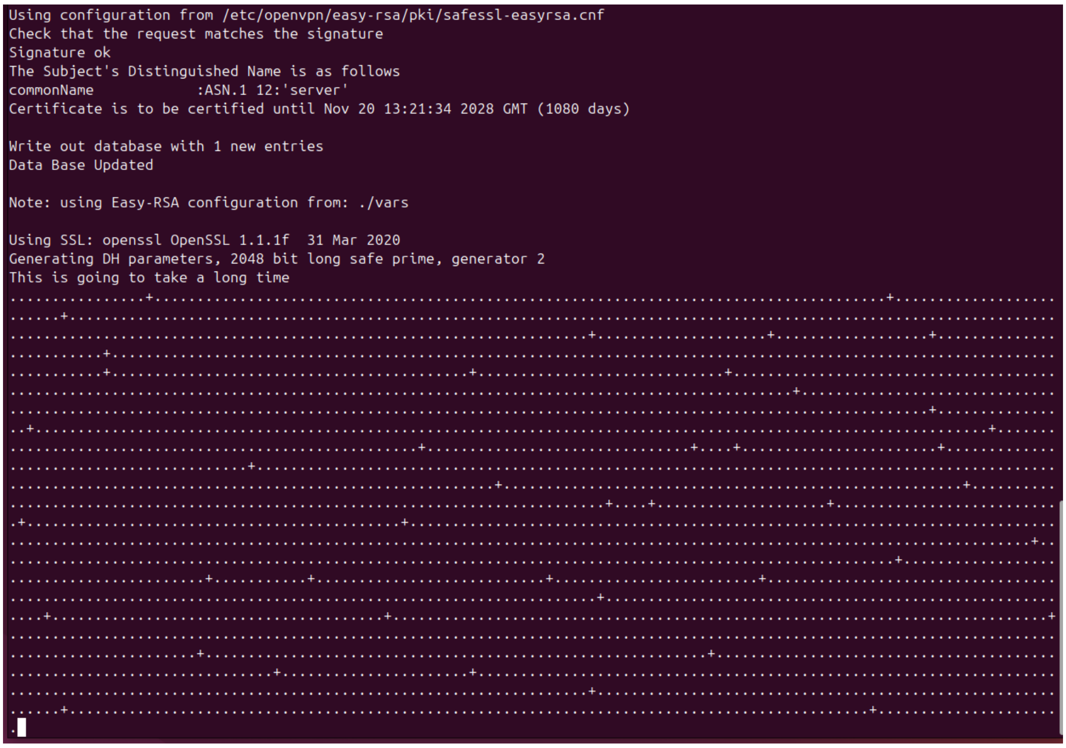

Figure 11: OpenVPN PKI generation and certificate/database initialization, the key generation takes longer depending on your machine resource.

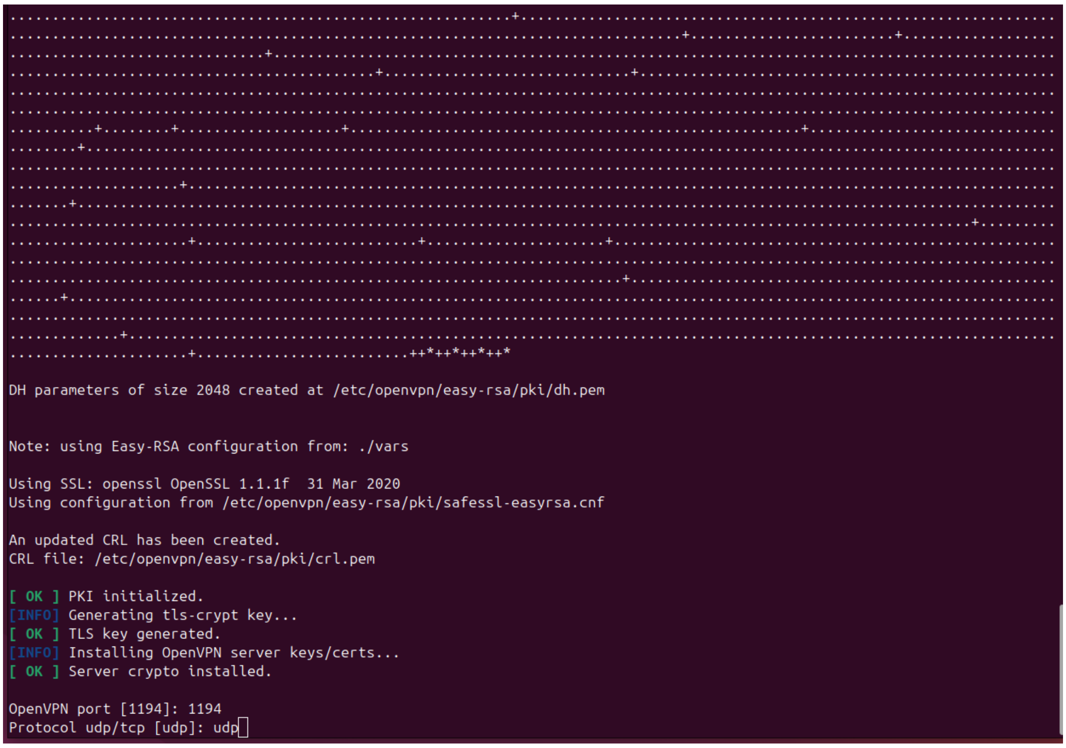

Figure 12: DH parameter generation and TLS crypt key production.

3.10 Successful Dual-VPN Completion

Once OpenVPN and WireGuard are installed, Bahdleen VPN applies NAT rules that cover both services and confirms that each system component has been configured successfully.

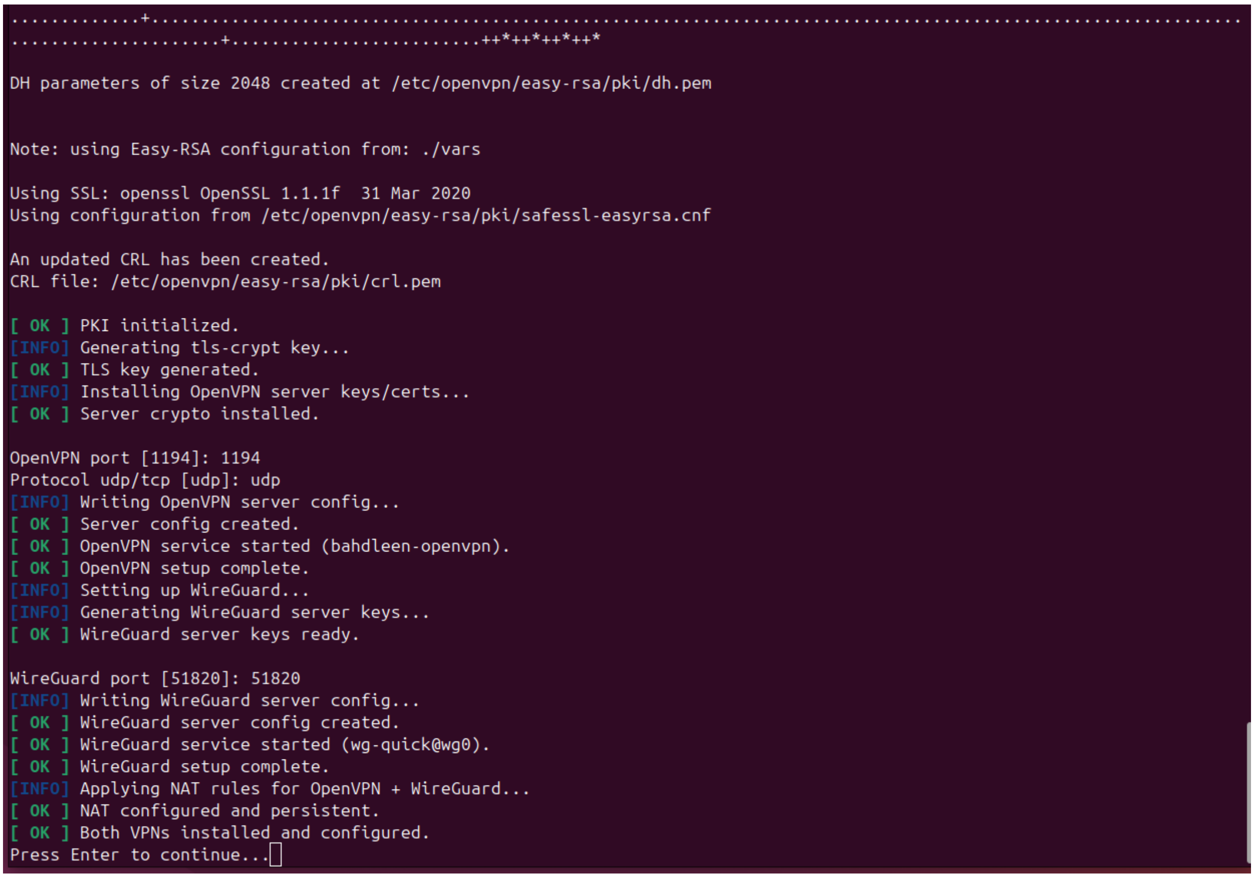

Figure 13: Combined setup completion showing OpenVPN + WireGuard configured and NAT applied.

3.11 Adding an OpenVPN Client Profile

Client provisioning is intentionally guided and flexible. The tool supports username input, certificate validity control, and optional passphrase protection for private keys.

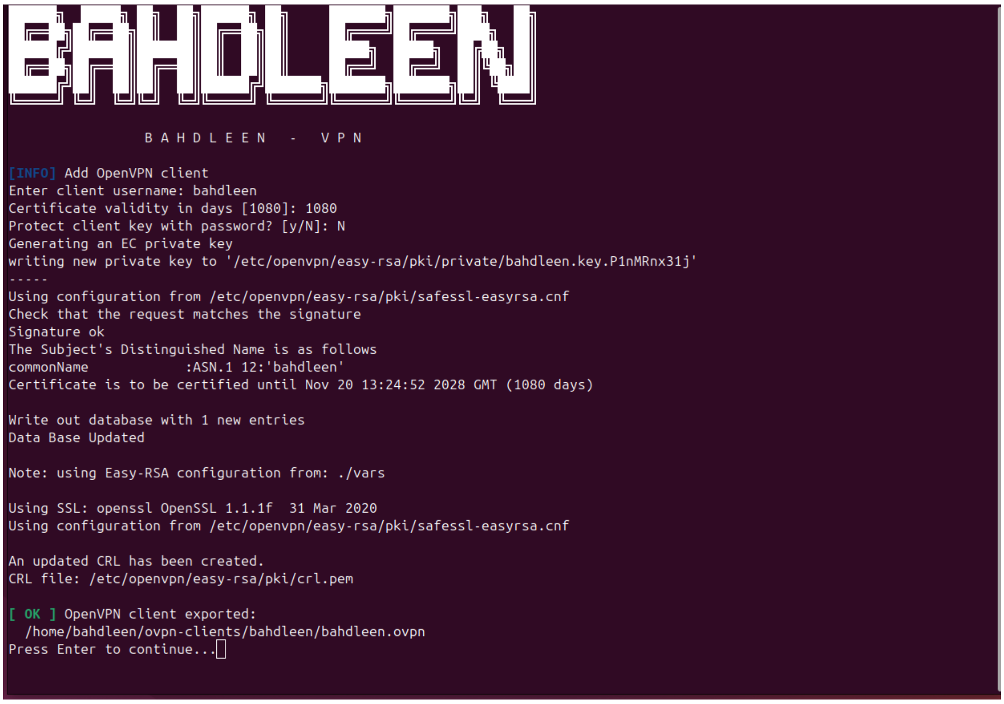

Figure 14: OpenVPN client creation prompt: username, certificate validity, and optional password protection.

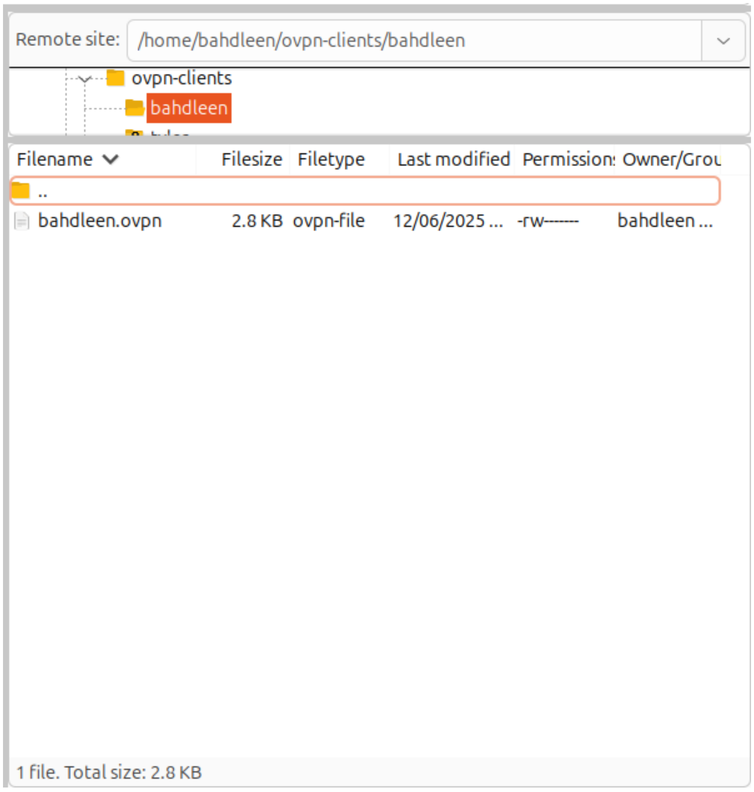

Figure 15: Successful export of the OpenVPN client configuration into ˜/`ovpn-clients`.

3.12 Reopening the Tool After Installation

After the global command is installed, Bahdleen VPN can be launched from anywhere on the system without returning to the project directory. This improves daily usability and makes the tool feel like a native utility rather than a one-time script.

From this point onward, `bahdleen-vpn` is the key command to start the tool.

```
bahdleen-vpn
```

With this workflow, administrators can repeatedly add or revoke clients without restarting the entire setup process, supporting practical operational usage.

This sequence forms a complete and reproducible installation narrative suitable for documentation, portfolio evidence, and a step-by-step video demonstration.